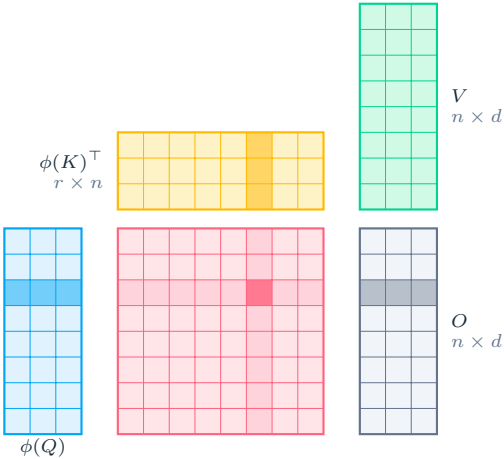
Pairs first: $(\phi(Q)\,\phi(K)^\top)\,V$

the only order open to softmax attention, whose e^x acts on each entry of $QK^\top$



$A = \phi(Q)\,\phi(K)^\top$
$A_{ij} = \phi(q_i)^\top \phi(k_j)$
$n \times n$, $2n^2 r$ operations

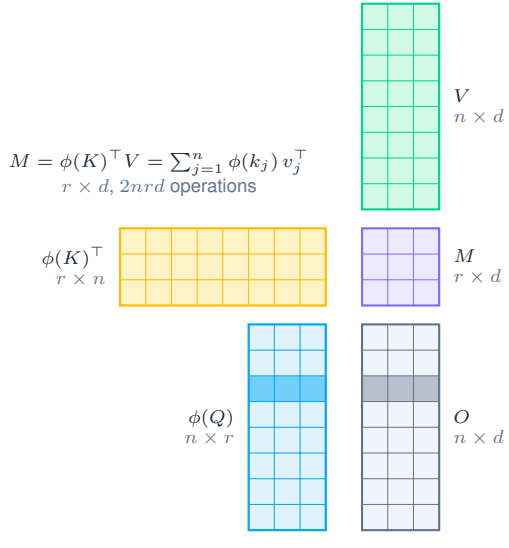
$O = AV$
$2n^2 d$ operations

$2n^2(r+d)$ **operations; holds** n^2 **numbers**

Summary first: $\phi(Q)\,(\phi(K)^\top V)$

open once the kernel is an inner product of features, $\phi(q)^\top \phi(k)$

$M = \phi(K)^\top V = \sum_{j=1}^{n} \phi(k_j)\, v_j^\top$
$r \times d$, $2nrd$ operations



$O = \phi(Q)\,M,\ o_i = M^\top \phi(q_i)$
$2nrd$ operations

$4nrd$ **operations; holds** rd **numbers**

Both orders give the same O, because matrix multiplication is associative. With $r = d$, the right order needs n/d times fewer operations.

Each row o_i is then divided by $\phi(q_i)^\top z$, where $z = \phi(K)^\top \mathbf{1}_n = \sum_j \phi(k_j)$ is formed once, in the same way.